

Foundation (Jalapeno)

- Q1.** What is the vertex form of a quadratic function?
(A) $y = a(x - h)^2 + k$
(B) $y = ax^2 + bx + c$
(C) $y = a(x + h)^2 - k$
(D) $y = x^2 + bx + c$
(E) $y = x^2 - bx - c$
- Q2.** A stock's price is modeled by $y = 2(x - 5)^2 + 10$. What is the vertex?
(A) (5, 10)
(B) (0, 0)
(C) (5, 0)
(D) (0, 10)
(E) (10, 5)
- Q3.** Write $y = x^2 + 4x + 4$ in vertex form.
(A) $y = (x + 2)^2$
(B) $y = (x - 2)^2$
(C) $y = (x + 2)^2 + 4$
(D) $y = (x - 2)^2 - 4$
(E) $y = x^2$
- Q4.** A savings account earns interest according to $y = 100(1.05)^x$. Write this in intercept form.
(A) $y = (0, 100)(1.05, 0)$
(B) $y = (100, 0)(0, 1.05)$
(C) $y = (0, 100)(1.05, 100)$
(D) $y = 100 + 1.05x$
(E) $y = (0, 100)(1.05, 100)$
- Q5.** What does the 'a' value represent in vertex form?
(A) x-coordinate of the vertex
(B) y-coordinate of the vertex
(C) width of the parabola
(D) direction of the parabola
(E) none of the above
- Q6.** Write $y = (x + 3)^2 - 2$ in standard form.
(A) $y = x^2 + 6x + 7$
(B) $y = x^2 + 6x - 2$
(C) $y = x^2 + 3x - 2$
(D) $y = x^2 - 6x + 2$
(E) $y = x^2 - 6x - 7$
- Q7.** The tax on a 10,000 investment is $y = 0.1(x - 5000)^2$. What is the tax when $x = 10000$?
(A) 0
(B) 500
(C) 1000
(D) 2500
(E) 5000
- Q8.** The vertex of $y = ax^2 + bx + c$ is $(-2, 3)$. Write the equation in vertex form.
(A) $y = a(x + 2)^2 + 3$
(B) $y = a(x - 2)^2 + 3$
(C) $y = a(x - 2)^2 - 3$
(D) $y = (x + 2)^2 + 3$
(E) $y = (x - 2)^2 - 3$

Open Ended Questions (FRQ)

- Q9.** Write $y = x^2 - 4x - 3$ in vertex and intercept form. Show your work and explain the significance of each form in the context of the stock market.
- Q10.** A company's profit is modeled by $y = 2(x - 10)^2 + 50$. Describe the profit when $x = 5$, $x = 10$, and $x = 15$. Explain the significance of the vertex in this context.

Chef's Tips

- Ensure students understand the vertex and intercept forms of quadratic equations.
- Provide examples of real-world applications, such as business and finance.
- Encourage students to show their work and explain their reasoning.